Executive Report

South African CPI Inflation Forecasting

EDAB6808

Assignment 2: Question 2

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Historical data: January 2021–December 2025
Forecast period: January–December 2026

An assessment of statistical and machine-learning forecasts of official headline year-on-year consumer price inflation, using chronological backtesting and an explicit discussion of forecast uncertainty.

1 Executive summary

The objective is to forecast South Africa's monthly headline year-on-year consumer price inflation for January–December 2026. The analysis uses 60 monthly observations from January 2021 to December 2025. The forecasting target is the official inflation rate in Statistics South Africa's Table B2; the supplied Table B1 CPI index is retained for data validation and exploratory analysis. This choice provides an interpretable percentage target and preserves the full required historical window.

The historically selected model is SARIMA$(1,1,0)(0,0,1)_{12}$, without deterministic drift. It achieved the lowest root mean squared error (RMSE) in the 2024 year-ahead selection exercise. Its advantage over the last-value naive benchmark was small: RMSE was 1.190 percentage points (pp), compared with 1.201 pp. The benchmark had the lower mean absolute error (MAE). The model choice therefore depends on the stated priority given to RMSE; it is not evidence that SARIMA dominates every alternative.

Table 1: Headline findings under comparable forecast designs.

Assessment	Finding
2024 year-ahead model selection	SARIMA has the lowest RMSE: 1.190 pp.
2025 year-ahead check, with choices fixed	Random Forest has the lowest RMSE: 0.321 pp; SARIMA: 0.416 pp.
2025 monthly one-step check	Last-value naive has the lowest RMSE: 0.242 pp; SARIMA has the lowest MAE: 0.179 pp.
Selected 2026 forecast	Mean of the twelve monthly year-on-year rates: 3.82%; December: 4.22%.
December 2026 uncertainty	Conditional 95% pointwise prediction interval: 0.55%–7.88%.

The statistical and Random Forest configurations were tuned using expanding historical origins ending in 2023. The model family was then selected using 2024 forecasts, before checking the fixed choices on 2025. All models forecast identical target months from identical origins. Random Forest's multi-step forecasts use earlier predictions when constructing later features, rather than using unavailable future observations. A separate monthly one-step evaluation shows how performance changes when fresh actual inflation observations become available.

The central 2026 path suggests moderately higher inflation towards year-end. It falls from 3.49% in January to 3.37% in February, then reaches approximately 4.22% in November and December. These are monthly year-on-year rates. Their 3.82% arithmetic mean is not the annual average CPI inflation rate calculated from annual-average index levels.

The forecast should be treated as a conditional planning baseline. The 95% prediction band widens substantially, and a residual diagnostic detects remaining serial dependence. Only five years of data are available; The models omit explanatory variables for future economic shocks. The different rankings across 2024, 2025 and forecast horizons further limit claims of a consistently superior model. The final SARIMA forecast retains the choice made before the 2025 check, while reporting the stronger 2025 Random Forest results transparently.

2 Data, target and preparation

2.1 Official source and consistent target

The data originate from Statistics South Africa's *Consumer Price Index, December 2025*, statistical release P0141. Table B1 reports the headline index numbers with December 2024 set to 100; Table B2 reports headline year-on-year rates for the same period. The modelling target is

$$\pi_t = 100\left(\frac{CPI_t}{CPI_{t-12}} - 1\right),$$

using the publisher's official Table B2 values, rather than recalculating the target from the rounded Table B1 index. A forecast of 4% means that the headline price index is expected to be 4% above its level in the same month of the preceding year. Forecast errors are measured in pp.

2.2 Monthly preparation and missing values

The supplied index file has five year rows, twelve month columns and an annual average. The month columns were reshaped into 60 dated observations, sorted chronologically and assigned a monthly frequency. Annual averages were excluded because they are summaries rather than additional months. The prepared index and official inflation target contain no missing months, duplicate dates or missing target values.

Calculating year-on-year inflation from a series starting in January 2021 would leave twelve undefined values: the corresponding 2020 index values are absent. These are structurally unavailable calculations, not unusual recorded inflation rates. Official Table B2 supplies the 2021 rates without inventing 2020 data or imputing missing targets. On the overlapping months, the largest absolute difference between the official rate and a calculation from rounded indices is 0.145 pp. One target definition is used throughout.

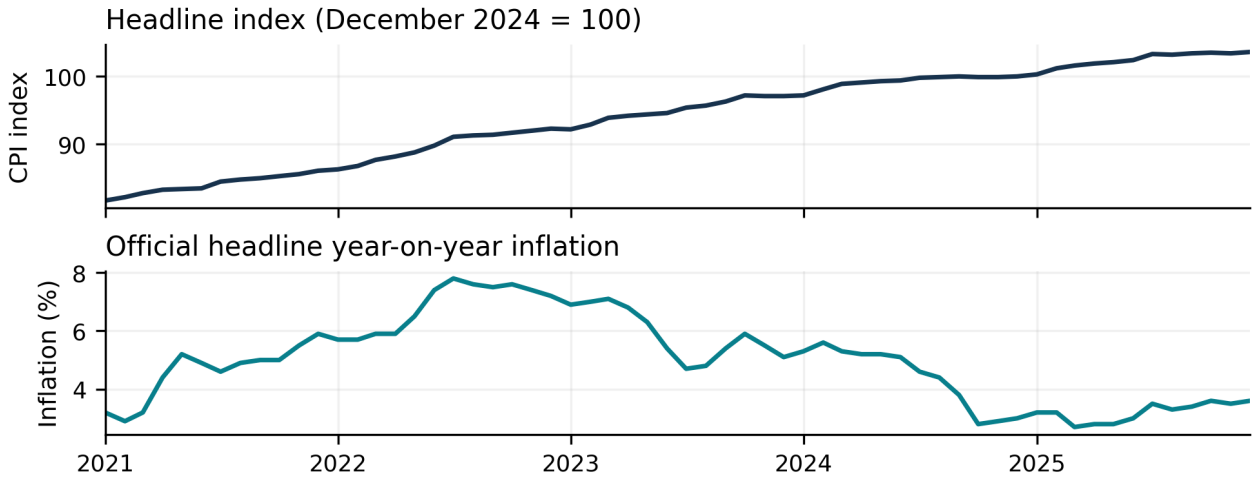


Figure 1: Headline CPI index and official year-on-year inflation, January 2021–December 2025. The index measures the price level; inflation measures its annual rate of change.

2.3 Transformations and information availability

No interpolation, outlier deletion, logarithmic transformation or scaling was applied to the target. First differencing is examined diagnostically and is part of the selected SARIMA specification. Random Forest uses past inflation and calendar features. No exogenous dataset is included.

The December 2025 observation was published on 21 January 2026. The final forecast is conditional on that release, rather than on information available on 31 December 2025. Historical checks use this publication vintage; they do not reconstruct archived release vintages or publication delays. No observed 2026 inflation is used to fit or assess the models.

3 Exploratory analysis and diagnostics

3.1 Trend, variability and seasonality

The monthly inflation rate averages 4.99% over the full sample, with a sample standard deviation of 1.51 pp. It peaks at 7.8% in July 2022 and reaches its minimum of 2.7% in March 2025. The index rises from 81.7 in January 2021 to 103.6 in December 2025, although inflation later declines: slower price growth does not imply that the price level has fallen.

Table 2: Descriptive statistics of monthly year-on-year inflation.

Year	Mean (%)	Minimum (%)	Maximum (%)	Std. dev. (pp)
2021	4.56	2.9	5.9	0.96
2022	6.85	5.7	7.8	0.84
2023	5.91	4.7	7.1	0.88
2024	4.43	2.8	5.6	1.04
2025	3.22	2.7	3.6	0.32

Means are arithmetic averages of monthly year-on-year rates, not annual-average CPI inflation.

An additive decomposition with period 12 separates the observed series into trend, seasonal and residual components. The changing trend dominates the estimated seasonal variation. With only five annual cycles, the seasonal pattern is tentative. The full-sample decomposition is descriptive only; its two-sided trend filter is never used as a predictive feature.

3.2 Stationarity and autocorrelation

Diagnostics use January 2021–December 2022, before the tuning outcomes. The augmented Dickey–Fuller (ADF) test has a unit-root null; the KPSS test has a level-stationarity null. The results favour considering first differencing, while the short sample limits the strength of this conclusion.

Table 3: Stationarity tests on the initial historical training window.

Series	Observations	ADF p-value	KPSS p-value
Inflation level	24	0.7902	0.0159
First difference	23	0.000037	> 0.10

The KPSS first-difference statistic is beyond the tabulated upper p-value range. Failure to reject a null does not prove it.

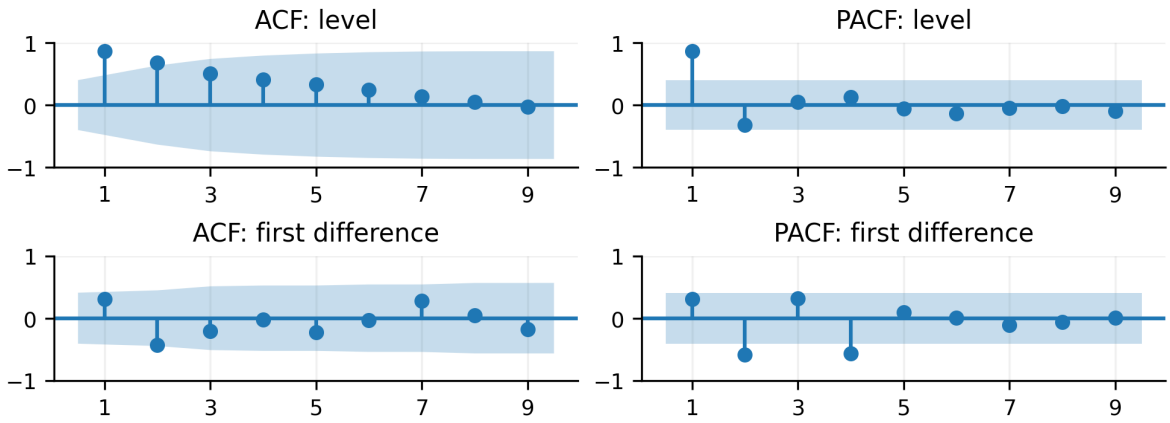


Figure 2: ACF and PACF of inflation levels and first differences in the initial training window. Persistence supports a small ARIMA/SARIMA search rather than a random split.

Adjacent year-on-year rates share overlapping price changes, so dependence is expected. The ACF/PACF guide a small candidate search; model orders and seasonal terms are ultimately assessed by historical forecast accuracy.

4 Methodology: chronological validation

4.1 Separate tuning, selection and evaluation

Forecasting is assessed by repeatedly predicting dates after the end of the available history. Training windows expand from January 2021; no random train/test split is used. The primary comparison predicts an entire twelve-month path without inserting the actual observations from inside that path. This matches the final task of forecasting all twelve months of 2026 from one historical endpoint.

Table 4: Chronological design and permitted use of each period.

Stage	Origin / forecast period	Purpose and restriction
Initial diagnostics	History through Dec. 2022	Inspect levels, differences, ACF/PACF and stationarity.
Configuration tuning	Dec. 2022; Mar., Jun., Sep. 2023	Predict only remaining months through Dec. 2023; horizons 12, 9, 6 and 3. Choose one configuration per family.
Family selection	Dec. 2023 / Jan.–Dec. 2024	Compare fixed configurations and benchmarks. Select on RMSE, with MAE breaking exact ties.
Locked year-ahead check	Dec. 2024 / Jan.–Dec. 2025	Evaluate fixed family and configurations; do not change the winner using this result.
Supplementary check	Dec. 2024–Nov. 2025 / next month	Refit with expanding available history; assess one-step forecasts separately.
Final refit	Dec. 2025 / Jan.–Dec. 2026	Use all 60 target observations and the previously selected model.

The four tuning origins yield 30 forecast-origin/horizon errors per configuration. Each error contributes equally to the pooled score, so earlier, longer paths contribute more errors. Some target months appear at multiple horizons; the tuning errors are therefore dependent. They guide configuration choice and are not presented as an independent test. The 2024 selection result also reflects a model-choice decision. The separate 2025 check provides evidence after that choice was fixed.

4.2 Common evaluation measures

For actual inflation y_i, forecast $\hat{y}_i$ and error $e_i = y_i - \hat{y}_i$, the primary measures are

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|e_i|, \qquad \text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}e_i^2}.$$

Both are expressed in pp. MAE describes typical absolute error; RMSE penalises large misses more strongly and is the stated selection criterion. Symmetric mean absolute percentage error (sMAPE) provides a supplementary relative measure:

$$\text{sMAPE} = \frac{100}{N}\sum_{i=1}^{N}\frac{2|y_i - \hat{y}_i|}{|y_i| + |\hat{y}_i|}.$$

Percentage measures can be unstable near zero, so they do not determine the winner. A zero actual and forecast contribute zero in the implementation. Mean absolute scaled error (MASE) divides each absolute error by the mean absolute one-step naive error in that origin's training history. This scale uses past data only. A value below one means error is below that historical scale; direct year-ahead benchmark comparisons remain necessary.

5 Statistical and machine-learning models

5.1 Statistical model: SARIMA

Eight compact ARIMA/SARIMA configurations were evaluated at the four tuning origins. The chosen specification is SARIMA$(1,1,0)(0,0,1)_{12}$ with no deterministic drift. It combines a first difference, one non-seasonal autoregressive term and one seasonal moving-average term at lag 12. Its pooled tuning RMSE is 0.890 pp. Historical forecasting accuracy determines the specification; the stationarity tests and ACF/PACF provide supporting diagnostics.

All candidate statistical fits converged at the tuning origins. Warnings about limited data for seasonal starting parameters were recorded rather than globally suppressed. The selected model also converged in the diagnostic fit through December 2024, with AIC 44.302. AIC is a within-sample fit criterion and does not replace backtest accuracy.

5.2 Machine-learning model: Random Forest

The Random Forest predicts the same inflation target from ten features: lags 1, 2, 3, 6 and 12; three- and six-month rolling means; six-month rolling standard deviation; and sine/cosine terms for the forecast month. Rolling features are shifted so that they exclude the target month. Calendar features are known in advance. Neither the contemporaneous CPI index nor the full-sample decomposition is used as a predictor.

Six configurations combine maximum tree depth 3 or unlimited depth with minimum leaf sizes 1, 2 or 3. Each uses 200 trees and all features at each split. The selected configuration has unlimited depth and minimum leaf size 2, with tuning RMSE 1.480 pp. The random seed is fixed at 42. The twelve-month lag leaves 48 usable supervised rows in the final fit. The small sample restricts how much nonlinear structure can be learned.

For a multi-step forecast, the forest is fitted using only the origin's history. It predicts the next month, appends that prediction, rebuilds features, and repeats. Later forecast features thus contain earlier forecasts. Errors can accumulate, and tree averaging has limited ability to extrapolate beyond training responses. Feature scaling is unnecessary for these tree splits.

5.3 Benchmarks and residual adequacy

The last-value naive model repeats the final observed inflation rate. The seasonal naive model repeats the rate from the same month of the previous year. These provide direct tests of whether model complexity improves forecasting. A volatility model is not required for this study's conditional-mean forecasting objective.

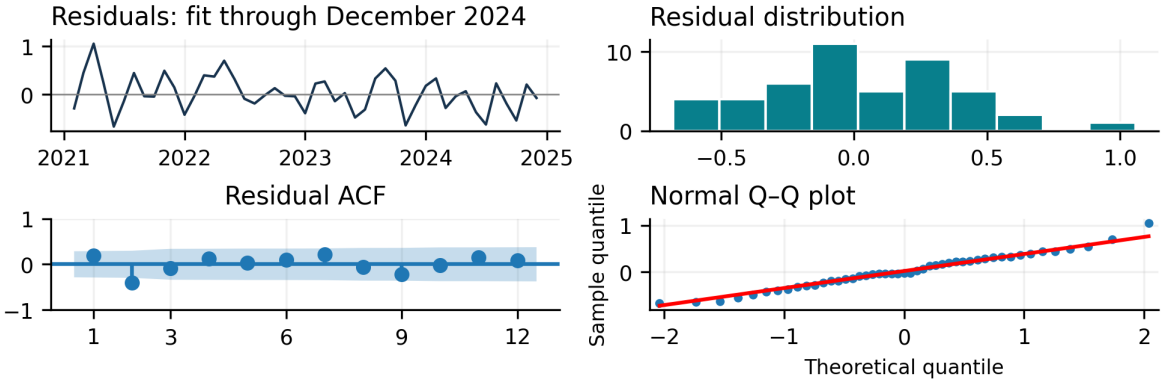


Figure 3: Selected SARIMA residuals from the fit through December 2024: time plot, distribution, ACF and Q–Q plot. Forty-seven residuals remain after the initial burn-in observation.

The lag-12 Ljung–Box test, adjusted for two AR/MA parameters, gives $p = 0.0296$. At the 5% level this indicates remaining serial dependence. The fitted model is usable for comparison but its residuals are not fully consistent with white noise. Gaussian prediction intervals should therefore be read as conditional approximations, with their limitations stated.

6 Results: common year-ahead comparisons

6.1 2024 selection exercise

All four methods forecast January–December 2024 from December 2023, using the same 36-month history and their historically chosen configurations. Each score summarises twelve forecasts from that single origin.

Table 5: 2024 year-ahead selection results. Lower values are better.

Model	MAE (pp)	RMSE (pp)	sMAPE (%)	MASE
SARIMA	0.943	**1.190**	21.90	2.642
Last-value naive	**0.850**	1.201	**20.02**	**2.380**
Seasonal naive	1.475	1.711	29.97	4.130
Random Forest	1.335	1.722	28.43	3.739

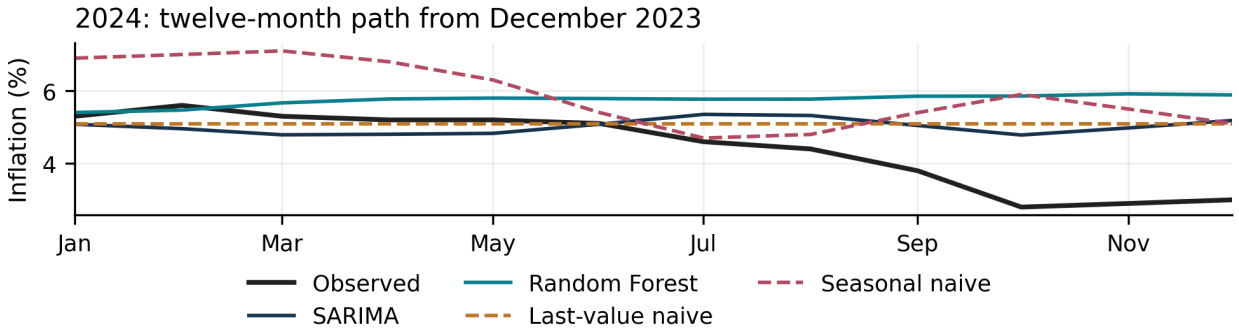


Figure 4: Observed 2024 inflation and forecasts made from December 2023. The late-year decline is difficult for all four methods to anticipate.

SARIMA wins the predefined RMSE criterion by only 0.011 pp relative to the last-value benchmark, approximately a 0.9% improvement. The benchmark wins on MAE and sMAPE. SARIMA's selection is consequently narrow and criterion-dependent. Random Forest does not improve on either benchmark's RMSE in this period. The high MASE values indicate that year-ahead errors exceed the historical one-step naive error scale.

6.2 2025 check with model choices fixed

The same configurations forecast January–December 2025 from December 2024, using the available 48-month history. The 2024 family selection remains fixed before these outcomes are inspected.

Table 6: 2025 year-ahead check results, without reselecting the model.

Model	MAE (pp)	RMSE (pp)	sMAPE (%)	MASE
Random Forest	**0.255**	**0.321**	**8.03**	**0.750**
Last-value naive	0.333	0.379	10.53	0.979
SARIMA	0.322	0.416	9.08	0.947
Seasonal naive	1.550	1.751	39.13	4.553

Random Forest performs best on every reported measure in this check; its RMSE is about 23% below SARIMA's. The naive benchmark also has a lower RMSE than SARIMA. Inflation is relatively stable in 2025, which may help explain the changed ranking, but the experiment does not establish the cause. SARIMA is retained for the final forecast to preserve the earlier selection decision. The result supports reporting Random Forest as a competitive alternative, rather than claiming universal SARIMA superiority. Each annual comparison has one origin and twelve dependent errors, so the differences are descriptive rather than statistically established improvements.

7 Forecast horizon and the 2026 outlook

7.1 Supplementary monthly one-step evaluation

The 2025 one-step check expands the history each month and refits the models before predicting the next observation. It therefore answers a different question from forecasting a full year without updated actuals.

Table 7: 2025 expanding-window one-step results.

Model	MAE (pp)	RMSE (pp)	sMAPE (%)	MASE
Last-value naive	0.183	**0.242**	5.79	0.564
SARIMA	**0.179**	0.253	**5.69**	**0.544**
Random Forest	0.194	0.255	6.16	0.595
Seasonal naive	1.550	1.751	39.13	4.710

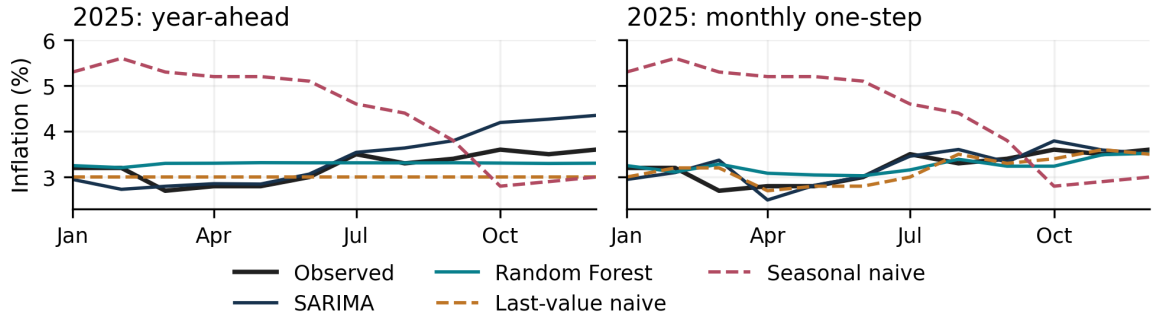


Figure 5: 2025 year-ahead forecasts (left) and monthly one-step forecasts (right). Fresh actual observations are incorporated only in the one-step design.

The last-value benchmark has the lowest one-step RMSE, while SARIMA has the lowest MAE. Their closeness shows the value of retaining simple benchmarks. The seasonal naive model performs poorly in both designs, so mechanically repeating the preceding year's rates is not supported. One-step accuracy should not be presented as evidence of equivalent twelve-step accuracy: long paths contain additional uncertainty and, for Random Forest, recursively generated features.

7.2 Refitting and the central forecast

Following the fixed selection, SARIMA is refitted to all 60 inflation observations from January 2021 to December 2025. The forecast starts in January 2026 and ends in December 2026. No 2026 target observations enter the model. The final workbook contains the required month and point forecast; the model's interval estimates are also reported here.

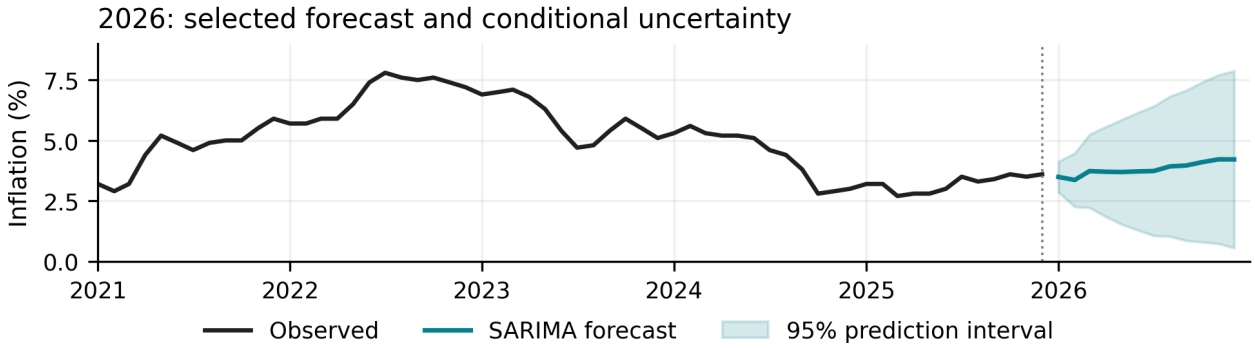


Figure 6: Selected SARIMA forecast for January–December 2026, with a conditional 95% pointwise prediction band and the complete historical target series.

8 Interpretation, uncertainty and conclusion

Table 8: Monthly 2026 forecasts and conditional 95% prediction limits.

Month	Point forecast (%)	Lower 95% (%)	Upper 95% (%)
January	3.49	2.85	4.13
February	3.37	2.25	4.48
March	3.73	2.22	5.24
April	3.70	1.84	5.56
May	3.69	1.53	5.85
June	3.72	1.29	6.15
July	3.73	1.06	6.41
August	3.93	1.03	6.83
September	3.96	0.85	7.07
October	4.10	0.80	7.40
November	4.22	0.73	7.71
December	4.22	0.55	7.88

Rounded to two decimals. Bounds are prediction intervals for individual monthly outcomes, not a simultaneous interval for the whole year or an interval for annual-average inflation.

8.1 Meaning and limitations of the uncertainty band

The interval widens from approximately 2.85%–4.13% in January to 0.55%–7.88% in December. This growing spread is more informative than the point forecast alone. The band follows the fitted statistical model's assumptions; its nominal 95% level has not been established by an empirical coverage study. It does not incorporate all parameter, model-selection, data-revision or future-shock uncertainty. Remaining residual dependence further weakens confidence in exact calibration.

Only five annual cycles support estimation. The 2025 CPI basket and weight changes introduce a comparability consideration, while the common December 2025 vintage does not reproduce real-time data releases. Exchange-rate changes, energy costs, administered prices and policy developments are not separately modelled. The analysis cannot attribute forecast movements to these drivers or assume that past dynamics persist.

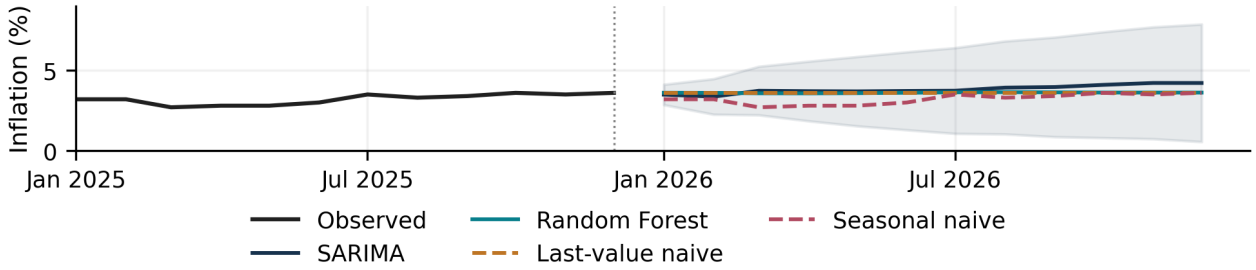


Figure 7: Alternative 2026 paths from the same historical endpoint. The shaded interval belongs to SARIMA only; the other models do not have calibrated prediction intervals in this analysis.

8.2 Conclusion and practical use

The study meets the forecasting objective through an official, consistent target; past-only tuning; common year-ahead comparisons; a separate 2025 check; and a full-history refit. SARIMA is the historically selected candidate, with a central December 2026 forecast of 4.22%. Random Forest's stronger 2025 result and the competitive naive benchmark show that the preferred approach remains sensitive to the period, horizon and loss measure. The report therefore supports a conditional baseline, not a claim of a reliably dominant model.

Forecasts should be reviewed as new releases become available, with the evaluation protocol fixed before judging revisions. Longer histories, correctly timed explanatory variables and interval-coverage checks would strengthen comparisons. Current planning should consider the widening prediction band and alternative paths alongside the central estimate.

Sources

The external data and methodological sources are listed below. All comparison tables, diagnostic summaries and figures in this report were produced from the executed Question 2 notebook and its saved results.

1. **Statistics South Africa.** (2026). *Consumer Price Index, December 2025*. Statistical release P0141, released 21 January 2026. Printed page 4, Table B1: CPI headline index numbers (December 2024=100), and Table B2: CPI headline year-on-year rates. Primary source for all 60 historical monthly observations.

   https://www.statssa.gov.za/publications/P0141/P0141December2025.pdf

2. **Hyndman, R. J., and Athanasopoulos, G.** (2021). *Forecasting: Principles and Practice*, third edition. OTexts, Melbourne. Section 5.10, "Time series cross-validation": expanding origins and assessment over multiple forecast horizons.

   https://otexts.com/fpp3/tscv.html

3. **Hyndman, R. J., and Athanasopoulos, G.** (2021). *Forecasting: Principles and Practice*, third edition. OTexts, Melbourne. Section 5.8, "Evaluating point forecast accuracy": forecast-error measures and scaled errors. The report uses the explicit sMAPE definition given in its methodology.

   https://otexts.com/fpp3/accuracy.html

4. **statsmodels documentation.** *SARIMAX*: seasonal ARIMA specification and state-space estimation. Supports the statistical implementation and its forecast distribution.

   https://www.statsmodels.org/stable/generated/statsmodels.tsa.statespace.sarimax.SARIMAX.html

5. **scikit-learn documentation.** *RandomForestRegressor*: regression forests, tree and leaf-size settings, and reproducible estimation. Supports the machine-learning implementation and parameter definitions.

   https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html

6. **South African Reserve Bank.** (2025). *The 2023 reweighted and rebased consumer price index*. Box 1, Quarterly Bulletin. Background on the CPI basket and weight changes introduced in 2025 and the linking of the historical series.

   https://www.resbank.co.za/content/dam/sarb/publications/quarterly-bulletins/boxes/2025/Box%201.pdf